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1. You are given a locked digital safe that opens only when a **specific 10-bit binary code** is entered. Each digit is 0 or 1. Your task is to find out the correct code using Genetic Algorithm. You have a MAGIC BOX. It knows the correct code is 0000011111, but of course you don't know. You will use this box for fitness calculation.

- Define an **encoding** scheme, and generate an **initial population of 4 chromosomes** using it. [2]
- Define a **fitness** function, and use it to calculate the **fitness** of the initial population. Think how you can use the MAGIC BOX. [2]
- Choose the **second** and **third** fittest chromosome as parents, and generate **TWO** children using **single-point crossover**. Use the last digit of your student ID as the **crossover index**. [2]
- Calculate the fitness of both children, and identify the **best solution** at this point. [2]

SOLUTION:

Encoding scheme: We choose a 10-bit binary code as the chromosome (candidate solution), where each bit (0 or 1) is chosen arbitrarily.

Fitness function: We define fitness function to be the number of bits of a chromosome that MATCH with the target. We do not know the target but the MAGIC BOX does (0000011111).

Initial population and Fitness:

SI	Chromosome	Fitness
P1	1010101010	4
P2	0000000000	5
P3	0001110011	6
P4	1111111111	5

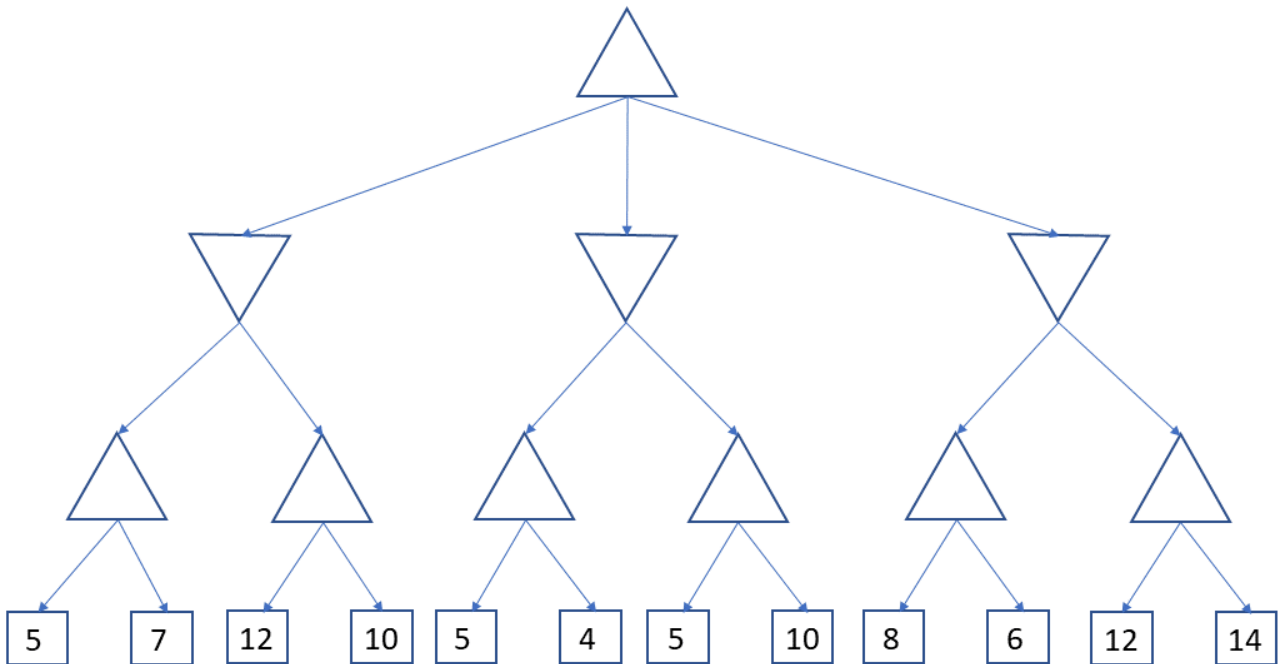
Fitness order of parents: $P3 > P2 = P4 > P1$.

Crossover:

Parent	Chromosome	Crossover Index	Children	Children Chromosome	Fitness of children
P2	0000000000	4	C1	0000011111	10
P4	1111111111		C2	1111100000	0

Best solution: C1 (0000011111), which in this case is the target.

2. Simulate **alpha-beta pruning** for the following game tree. For each node, write the minimax value inside the node, and write the values of alpha and beta beside the node. Also show which branches and nodes will be pruned by putting a cross mark on/beside them. [7]



SOLUTION:

